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Wire harness

Abstract

A wire harness including: an electric wire; an exterior tube that covers an outer periphery of the electric wire; a path regulator that extends along a lengthwise direction of the exterior tube while covering a portion in a circumferential direction of an outer periphery of the exterior tube, and regulates a path along which the electric wire is routed; and a fixing member that holds the path regulator and is fixed to a vehicle body.

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Background/Summary

BACKGROUND

(1) The present disclosure relates to a wire harness.

(2) Conventionally, a wire harness that includes a corrugated tube that covers the outer periphery of an electric wire member, and a path regulating member that covers a portion in the circumferential direction of the corrugated tube and regulates a path along which the electric wire member is routed is known (for example, see JP 2013-55760A).

(3) The corrugated tube of the wire harness described in JP 2013-55760A includes a slit formed along the lengthwise direction thereof. The path regulating member includes a path retaining member provided along the outer periphery of the corrugated tube, and an attachment member provided in the slit. The attachment member is configured to be engageable with the inner peripheral side portion of the slit and the outer peripheral side portion of the path retaining member. The corrugated tube, the path retaining member, and the attachment member are fixed together by, for example, winding a piece of tape around them, and the path of the electric wire member is thereby regulated.

(4) In addition, a configuration is conceivable in which, for example, a belt clamp is wound around and collectively holds the integrated corrugated tube and path regulating member, and is fixed to a vehicle body, thereby fixing the wire harness to the vehicle body.

SUMMARY

(5) However, for example, in a configuration in which a path regulating member is held by a fixing member such as a belt clamp as described above, there is a risk that the path regulating member will rotate in the circumferential direction relative to the fixing member. Moreover, rotation of the path regulating member relative to the vehicle body can cause a section of the corrugated tube that is not covered by the path regulating member to be exposed to the outside, for example, to oppose the ground, and accordingly, can cause a reduction in the durability of the wire harness.

(6) An exemplary aspect of the disclosure provides a wire harness capable of suppressing rotation of a path regulating member relative to a fixing member that is fixed to a vehicle body.

(7) A wire harness according to the present disclosure includes: an electric wire; an exterior tube that covers an outer periphery of the electric wire; a path regulator that extends along a lengthwise direction of the exterior tube while covering a portion in a circumferential direction of an outer periphery of the exterior tube, and regulates a path along which the electric wire is routed; and a fixing member that holds the path regulator and is fixed to a vehicle body, wherein the path regulator includes: an insertion port that is an opening that extends along a lengthwise direction of the path regulator over an entire length of the path regulator and is configured to enable the exterior tube to be inserted therein, and a first end and a second end that are positioned on opposite sides to each other in a circumferential direction of the path regulator, and from the insertion port, the fixing member includes an engagement that protrudes toward the insertion port, and the engagement is capable of coming into contact with at least one of the first end and the second end in the circumferential direction of the path regulator.

(8) With the wire harness according to the present disclosure, it is possible to suppress rotation of a path regulator relative to a fixing member that is fixed to a vehicle body.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic diagram of a configuration showing a wire harness according to an embodiment of the present disclosure.

(2) FIG. 2 is a cross-sectional view showing a portion of the wire harness according to the embodiment.

(3) FIG. 3 is a side view of the wire harness according to the embodiment when fixed to a vehicle body.

(4) FIG. 4 is an exploded perspective view of a portion of the wire harness according to the embodiment.

(5) FIG. 5 is a front view showing a portion of the wire harness according to the embodiment.

(6) FIG. 6 is a cross-sectional view taken along the line 6-6 in FIG. 5.

DETAILED DESCRIPTION OF EMBODIMENTS

Description of Embodiments of Disclosure

(7) First, embodiments of the present disclosure will be listed and described. [1] A wire harness according to the present disclosure includes: an electric wire member, a tubular exterior member that covers an outer periphery of the electric wire member, a path regulating member that extends along a lengthwise direction of the exterior member while covering a portion in a circumferential direction of the outer periphery of the exterior member, and regulates a path along which the electric wire member is routed, and a fixing member that holds the path regulating member and is fixed to a vehicle body, the path regulating member includes: an insertion port that is an opening that extends along the lengthwise direction of the path regulating member over the entire length of the path regulating member and is configured to enable the exterior member to be inserted therinto, and a first end portion and a second end portion that are positioned on opposite sides to each other in the circumferential direction of the path regulating member, and form the insertion port, the fixing member includes an engagement portion that protrudes toward the insertion port, and the engagement portion is capable of coming into contact with at least one of the first end portion and the second end portion in the circumferential direction of the path regulating member.

(8) According to the above configuration, the path regulating member can be retrofitted to the outer periphery of the exterior member through the insertion port. The engagement portion of the fixing member can come into contact with at least one of the first end portion and the second end portion that form the insertion port in the circumferential direction of the path regulating member, and thus the fixing member can keep the path regulating member from rotating relative to the fixing member in the circumferential direction, while holding the path regulating member. [2] Preferably, the engagement portion includes a narrow portion whose width decreases toward a leading end side of the engagement portion when viewed from the lengthwise direction of the path regulating member.

(9) According to the above configuration, the engagement portion includes the narrow portion whose width decreases toward the leading end side of the engagement portion when viewed from the lengthwise direction of the path regulating member, and thus the engagement portion can be easily inserted into the insertion port, which facilitates attachment. [3] Preferably, the fixing member includes a main body portion and a lid portion that is fixable to the main body portion, and encloses and holds the path regulating member in cooperation with the main body portion while fixed to the main body portion.

(10) According to the above configuration, the fixing member includes the main body portion and the lid portion that is fixable to the main body portion and encloses and holds the path regulating member in cooperation with the main body portion while fixed to the main body portion, and thus the fixing member can be retrofitted to the path regulating member. [4] Preferably, the engagement portion protrudes from a vehicle body opposing portion that opposes an attachment surface of the vehicle body.

(11) According to the above configuration, the engagement portion protrudes from the vehicle body opposing portion that opposes the attachment surface of the vehicle body, and thus the insertion port of the path regulating member faces the attachment surface side. Thus, for example, the insertion port can be kept from facing downward so as to oppose the ground. [5] Preferably, a plurality of fixing members are provided in the lengthwise direction of the path regulating member.

(12) According to the above configuration, since a plurality of fixing members are provided in the lengthwise direction of the path regulating member, rotation of the path regulating member relative to the fixing members in the circumferential direction can be suppressed at a plurality of locations in the lengthwise direction of the path regulating member. Thus, it is possible to more firmly suppress rotation of the path regulating member. [6] Preferably, the exterior member is a corrugated tube that has a bellows structure in which an annular protrusion and an annular recess are alternately provided in a continuous manner along the lengthwise direction of the exterior member, and the engagement portion includes a protrusion that enters the annular recess.

(13) According to the above configuration, the engagement portion includes the protrusion that enters the annular recess of the corrugated tube, and thus it is possible to suppress movement of the corrugated tube relative to the fixing member in the lengthwise direction of the corrugated tube. [7] Preferably, the wire harness further includes a slide regulating member that regulates movement of the path regulating member relative to the exterior member in the lengthwise direction of the exterior member, and the path regulating member is allowed to move relative to the fixing member in the lengthwise direction of the exterior member in a case where the slide regulating member is not attached.

(14) According to the above configuration, the slide regulating member regulates movement of the path regulating member relative to the exterior member in the lengthwise direction of the exterior member. In addition, the path regulating member is allowed to move relative to the fixing member in the lengthwise direction of the exterior member in a state where no slide regulating member is attached, and thus it is possible to adjust the position of the path regulating member. [8] Preferably, the path regulating member includes a protruding portion that protrudes from an inner surface of at least one of the first end portion and the second end portion toward the exterior member, and comes into contact with an outer surface of the exterior member.

(15) According to the above configuration, the path regulating member includes the protruding portion that protrudes from the inner surface of at least one of the first end portion and the second end portion and comes into contact with the outer surface of the exterior member, and thus it is possible to keep the path regulating member from coming loose from the exterior member through the insertion port. [9] Preferably, the protruding portion protrudes from an inner surface of at least one of a leading end of the first end portion and a leading end of the second end portion.

(16) According to the above configuration, for example, compared with a case where the protruding portion protrudes from the inner surface of the path regulating member at a position slightly away from the leading end of the first end portion or the leading end of the second end portion, it is possible to further keep the path regulating member from coming loose from the exterior member through the insertion port.

DESCRIPTION OF EMBODIMENTS OF DISCLOSURE

(17) Specific examples of a wire harness according to the present disclosure will be described below with reference to the drawings. In the drawings, part of a structure may be exaggerated or simplified for convenience of explanation. In addition, the dimensional ratio of each part may differ between drawings. The present disclosure is not limited to these examples, but is defined by the claims and intended to include all modifications within the meaning and scope equivalent to the claims. The term “orthogonal” as used herein includes not only being exactly orthogonal but also being substantially orthogonal within the range in which the functions and effects according to the embodiment are achieved.

(18) Overall Configuration of Wire Harness **10**

(19) A wire harness **10** shown in FIG. **1** electrically connects two or three or more electric devices. The wire harness **10** electrically connects an inverter **11** installed on the front side of a vehicle **V** such as a hybrid vehicle or an electric vehicle and a high-voltage battery **12** installed rearward of the inverter **11** in the vehicle **V**, for example. The wire harness **10** is, for example, routed so as to pass under the floor of the vehicle **V**. For example, the wire harness **10** is routed so that an

intermediate portion of the wire harness in the longitudinal direction thereof extends outside the vehicle interior such as under the floor of the vehicle V.

(20) The inverter **11** is connected to a wheel driving motor (not shown) that is a motive power source for vehicle travel. The inverter **11** generates AC power from DC power of the high-voltage battery **12**, and supplies the resultant AC power to the motor. The high-voltage battery **12** is, for example, a battery capable of supplying a voltage of several hundred volts.

(21) As shown in FIGS. **1** and **2**, the wire harness **10** includes an electric wire member **20** (electric wire) that electrically connects the above electric devices, a tubular exterior member **30** (exterior tube) that covers the outer periphery of the electric wire member **20**, and a path regulating member **40** (path regulator) that covers the outer periphery of the exterior member **30** and regulates a path (hereinafter, referred to as a “routing path”) along which the electric wire member **20** is routed.

[0029] In addition, as shown in FIG. **3**, the wire harness **10** includes fixing members **50** that hold the path regulating member **40**, and are fixed to a vehicle body V1 of the vehicle V. A pair of connectors C1 and C2 are respectively attached to two end portions of the electric wire member **20**.

(22) Configuration of Electric Wire Member **20**

(23) The electric wire member **20** includes one or more electric wires **21** and a braided member **24** that collectively covers the outer peripheries of the electric wires **21**. The electric wire member **20** according to the present embodiment includes two electric wires **21**. One end portion of the electric wire member **20** is connected to the inverter **11** via the connector C1, and the other end portion of the electric wire member **20** is connected to the high-voltage battery **12** via the connector C2. The electric wire member **20** is formed in an elongated shape so as to extend in the front-rear direction of the vehicle, for example. The electric wires **21** are high-voltage electric wires that can support a high voltage/large current, for example. Each electric wire **21** may be a non-shielded electric wire that does not have an electromagnetic shield structure, or a shielded electric wire that has an electromagnetic shield structure, for example.

(24) Configuration of Electric Wires **21**

(25) As shown in FIG. **2**, the electric wires **21** are coated electric wires that each include a core wire **22** made of a conductor and an insulation coating **23** that covers the outer periphery of the core wire **22**.

(26) Configuration of Core Wire **22**

(27) A twisted wire formed by twisting a plurality of metal strands together, a columnar conductor made of a single columnar metal bar that has a solid structure, a tubular conductor that has a hollow structure, or the like can be used as the core wire **22**, for example. In addition, for example, a combination of a plurality of types of conductors such as a twisted wire, a columnar conductor, a tubular conductor, and the like can also be used as the core wire **22**. Examples of the columnar conductor can include a single core wire, a busbar, and the like. The core wire **22** according to the present embodiment is a twisted wire. A metal material that is copper-based, aluminum-based, or the like can be used as the material of the core wire **22**, for example.

(28) A cross-sectional shape (hereinafter, referred to as a “transverse cross-sectional shape”) of the core wire **22** taken along a plane orthogonal to the lengthwise direction of the core wire **22**, in other words the lengthwise direction of the electric wire **21** can have any shape. The transverse cross-sectional shape of the core wire **22** has a circular shape, a semicircular shape, a polygonal shape, a square shape, a flat shape, or the like. The transverse cross-sectional shape of the core wire **22** according to the present embodiment is circular.

(29) Configuration of Insulation Coating **23**

(30) The insulation coating **23** covers the entire outer circumference of the outer peripheral surface of the core wire **22**, for example. The insulation coating **23** is made of an insulating material such as a synthetic resin. A synthetic resin whose main component is a polyolefin-based resin such as cross-linked polyethylene or cross-linked polypropylene can be used as the material of the insulation coating **23**. Also, as the material of the insulation coating **23**, one kind of material can be

used, or two or more kinds of materials can be used in combination as appropriate.

(31) Configuration of Braided Member 24

(32) The braided member **24** is formed in a tubular shape so as to collectively cover the outer peripheries of the electric wires **21** as a whole, for example. The braided member **24** is provided so as to cover the outer peripheries of the electric wires **21** in the lengthwise direction of the electric wires **21** over substantially the entire length thereof, for example. A braided wire formed by braiding a plurality of metal strands, or a braided wire formed by braiding a metal strand and a resin strand in combination can be used as the braided member **24**. A metal material that is copper-based, aluminum-based, or the like can be used as the material of the metal strand, for example. Although not illustrated, the braided member **24** is grounded at the connector **C1** or **C2**, or the like.

(33) Configuration of Exterior Member 30

(34) As shown in FIG. **3**, the exterior member **30** is formed in a cylindrical shape so as to cover the outer periphery of the electric wire member **20** over the entire circumferential direction thereof. The exterior member **30** is sealed entirely in the circumferential direction. The exterior member **30** is provided so as to cover the outer periphery of a portion in the lengthwise direction of the electric wire member **20**, for example. The exterior member **30** according to the present embodiment is a corrugated tube that has a bellows structure in which annular protrusions **31** and annular recesses **32** are alternately provided in a continuous manner along the lengthwise direction of the exterior member **30**. The exterior member **30** is flexible.

(35) A conductive resin material or a nonconductive resin material can be used as the material of the exterior member **30**, for example. A synthetic resin such as polyolefin, polyamide, polyester, or an ABS resin can be used as the resin material.

(36) Configuration of Path Regulating Member 40

(37) As shown in FIGS. **2** and **4**, the path regulating member **40** covers a portion in the circumferential direction of the outer periphery of the exterior member **30** and extends along the lengthwise direction of the exterior member **30**. Note that the path regulating member **40** covers an area that is larger than half of the outer periphery of the exterior member **30**. The path regulating member **40** according to the present embodiment is attached to the outer periphery of a portion of the routing path of the electric wire member where the exterior member **30** extends linearly, such as a portion under the floor of the vehicle **V**.

(38) The path regulating member **40** is made of resin. A synthetic resin such as polypropylene, polyamide, or polyacetal can be used as the material of the path regulating member **40**. The path regulating member **40** can be manufactured using a known manufacturing method such as extrusion molding or injection molding. The cross-sectional shape of the path regulating member **40** according to the present embodiment is constant when viewed from the lengthwise direction thereof. The path regulating member **40** is an extrusion-molded component.

(39) The path regulating member **40** includes an insertion port **44** that is an opening extending in the lengthwise direction of the path regulating member **40**, a first end portion **41** and a second end portion **42** that form the insertion port **44**, and a coupling portion **43** that couples the first end portion **41** and the second end portion **42**. In other words, the path regulating member **40** includes the coupling portion **43** formed so as to cover a portion in the circumferential direction of the exterior member **30**, the first end portion **41** and the second end portion **42** provided at two end portions of the coupling portion **43**, and the insertion port **44** that is formed by the first end portion **41** and the second end portion **42**.

(40) The first end portion **41** and the second end portion **42** are positioned on opposite sides to each other in the circumferential direction of the path regulating member **40**. The first end portion **41** and the second end portion **42** are spaced apart from each other, sandwiching the insertion port **44** in the circumferential direction of the path regulating member **40**. The transverse cross-sectional shape of the first end portion **41**, the second end portion **42**, and the coupling portion **43** is an arc shape.

(41) The path regulating member **40** includes a pair of protruding portions **45** (protrusions) that protrude toward the exterior member **30** inserted into the path regulating member **40**, and come into contact with the outer surface of the exterior member **30**, more specifically, the outer surfaces of annular protrusions **31**. One of the pair of protruding portions **45** protrudes from the inner surface of the first end portion **41**. The other of the pair of protruding portions **45** protrudes from the inner surface of the second end portion **42**. The protruding portions **45** protrude from the inner surfaces of a leading end **46** of the first end portion **41** and a leading end **47** of the second end portion **42**. The transverse cross-sectional shape of each protruding portion **45** is semicircular, for example. The protruding portions **45** extend along the lengthwise direction of the path regulating member **40** over the entire length thereof.

(42) The leading end **46** of the first end portion **41** and the leading end **47** of the second end portion **42** are curved as viewed from the lengthwise direction of the path regulating member **40**. In other words, the transverse cross-sectional shape of each of the leading end **46** of the first end portion **41** and the leading end **47** of the second end portion **42** is curved. Specifically, the transverse cross-sectional shape of each of the leading end **46** of the first end portion **41** and the leading end **47** of the second end portion **42** is semicircular.

(43) The insertion port **44** extends along the lengthwise direction of the path regulating member **40** over the entire length thereof. The opening width of the insertion port **44**, that is to say, the shortest distance between the first end portion **41** and the second end portion **42** is smaller than the outer diameter of the exterior member **30**.

(44) When the exterior member **30** is inserted into the insertion port **44** from a direction orthogonal to the lengthwise direction, the path regulating member **40** elastically deforms, thus increasing the opening width of the insertion port **44**. Once the exterior member **30** is inserted into the path regulating member **40**, the path regulating member **40** elastically deforms so as to return to its original shape. Accordingly, the above opening width decreases to a width that is smaller than the outer diameter of the exterior member **30**, and thus the path regulating member **40** is attached to the exterior member **30**.

(45) Configuration of Fixing Member **50**

(46) As shown in FIG. 3, the fixing members **50** hold the path regulating member **40** attached to the exterior member **30**, and are fixed to an attachment surface **V2** of the vehicle body **V1**. The attachment surface **V2** according to the present embodiment is a surface outside the vehicle interior under the floor of the vehicle **V**, and is a surface opposing the ground. A plurality of fixing members **50** are provided in the lengthwise direction of the path regulating member **40**. In the present embodiment, two fixing members **50** are provided in the lengthwise direction of the path regulating member **40**.

(47) As shown in FIGS. 3 to 5, each of the fixing members **50** is made of resin. The fixing member **50** includes a main body portion **51** (main body) and a lid portion **52** (lid). The lid portion **52** can be fixed to the main body portion **51**, and can enclose and hold the path regulating member **40** in cooperation with the main body portion **51** while fixed to the main body portion **51**.

(48) To be specific, as shown in FIG. 5, the main body portion **51** includes an arc portion **53** formed in an arc shape so as to cover a portion of the outer periphery of the path regulating member **40** as viewed from the lengthwise direction of the path regulating member **40**, and a fixing portion **54** provided at one end of the arc portion **53**.

(49) The fixing portion **54** is formed in a substantially quadratic prism shape. As shown in FIGS. 3 and 4, the fixing portion **54** includes a fixing hole **55** that extends therethrough along a direction orthogonal to the attachment surface **V2**. The fixing portion **54** is also provided with a claw engagement portion **56**.

(50) As shown in FIG. 5, the lid portion **52** includes a lid arc portion **57** formed in an arc shape so as to cover a portion of the outer periphery of the path regulating member **40** as viewed from the lengthwise direction of the path regulating member **40**, and an extension portion **58** provided at one

end of the lid arc portion 57.

(51) The lid portion 52 is coupled to the main body portion 51 via a hinge portion 59. Specifically, the other end of the arc portion 53 of the main body portion 51 is coupled to the other end of the lid arc portion 57 of the lid portion 52 via the hinge portion 59. The hinge portion 59 according to the present embodiment is a thin portion. The main body portion 51 and the lid portion 52 are an integrally-molded component integrally formed via the hinge portion 59. The main body portion 51 and the lid portion 52 can rotate relative to each other about the hinge portion 59.

(52) As a result of being rotated relative to the main body portion 51 and brought into a closed state, the lid portion 52 can enclose the path regulating member 40 in cooperation with the main body portion 51. In addition, by rotating the lid portion 52 relative to the main body portion 51 and bringing it into an open state, the path regulating member 40 can be disposed between the lid portion 52 and the main body portion 51 from the outside. A claw portion 60 that can be engaged with the claw engagement portion 56 of the main body portion 51 is provided on the extension portion 58 of the lid portion 52. When the lid portion 52 is brought into the closed state, the claw portion 60 is engaged with the claw engagement portion 56. When the lid portion 52 is fixed to the main body portion 51 as a result of the claw portion 60 being engaged with the claw engagement portion 56 in this manner, the lid portion 52 can enclose and hold the path regulating member 40 in cooperation with the main body portion 51. In addition, as shown in FIGS. 3 and 4, a communication hole 61 that is in communication with the fixing hole 55 of the fixing portion 54 in the closed state is provided in the extension portion 58.

(53) As shown in FIG. 3, the fixing members 50 are fixed to the vehicle body V1 in a state where the lid portions 52 are in contact with the attachment surface V2 of the vehicle body V1. Fixing pieces 62 that extend in a direction orthogonal to the attachment surface V2, that is, toward the ground are provided on the attachment surface V2, for example. The fixing members 50 are then fixed to the attachment surface V2 as a result of the fixing piece 62 being passed through the communication hole 61 and the fixing hole 55 and engaging with an edge portion of the fixing hole 55 or the like.

(54) As shown in FIG. 5, the fixing member 50 includes an engagement portion 63 (engagement) that protrudes toward the insertion port 44 of the held path regulating member 40. In other words, the fixing member 50 includes the engagement portion 63 that protrudes so as to be disposed inside the insertion port 44 of the held path regulating member 40. The engagement portion 63 can come into contact with the first end portion 41 and the second end portion 42 in the circumferential direction of the path regulating member 40. The engagement portion 63 is disposed between the first end portion 41 and the second end portion 42 in the circumferential direction of the path regulating member 40, and, when the path regulating member 40 attempts to rotate relative to the fixing member 50 in the circumferential direction, the engagement portion 63 comes into contact with the first end portion 41 or the second end portion 42, and regulates rotation of the path regulating member 40.

(55) The engagement portion 63 is provided on the lid portion 52. Specifically, the engagement portion 63 protrudes from a vehicle body opposing portion 64 of the lid portion 52 that opposes the attachment surface V2 of the vehicle body V1, toward the insertion port 44. In other words, the engagement portion 63 protrudes from the attachment surface V2 side of the vehicle body V1. The engagement portion 63 protrudes toward the ground. Accordingly, the insertion port 44 of the path regulating member 40 faces upward, that is, the attachment surface V2 side.

(56) As shown in FIG. 6, the engagement portion 63 includes protrusions 65 that enter annular recesses 32 of the exterior member 30. The protrusions 65 protrude from the engagement portion 63. A plurality of protrusions 65 are provided in the lengthwise direction of the exterior member 30. In the present embodiment, five protrusions 65 are provided in the lengthwise direction of the exterior member 30, and correspondingly enter annular recesses 32 of the exterior member 30. Accordingly, movement of the exterior member 30 relative to the fixing member 50 in the

lengthwise direction of the exterior member **30** is suppressed.

(57) As shown in FIG. 5, the engagement portion **63** includes a narrow portion **66** whose width decreases toward a leading end side thereof when viewed from the lengthwise direction of the path regulating member **40**. Each protrusion **65** is provided with a narrow portion **66** according to the present embodiment. In other words, when viewed from the lengthwise direction of the path regulating member **40**, the width of the protrusion **65** decreases toward the leading end side thereof so as to form the narrow portion **66**.

(58) As shown in FIG. 3, the wire harness **10** includes slide regulating members **70** (slide regulators) for regulating movement of the path regulating member **40** relative to the exterior member **30** in the lengthwise direction of the exterior member **30**. The slide regulating members **70** according to the present embodiment are pieces of adhesive tape. The slide regulating members **70** regulate movement of the path regulating member **40** relative to the exterior member **30** by being wound from the path regulating member **40** to the exterior member **30**, respectively at the two ends of the path regulating member **40**. When the slide regulating members **70** are not attached, the path regulating member **40** is allowed to move relative to the fixing members **50** in the lengthwise direction of the exterior member **30**. That is to say, the fixing members **50** and the path regulating member **40** have no sections that are directly engaged with each other in the lengthwise direction of the path regulating member **40**. Thus, the position of the path regulating member **40** relative to the fixing members **50** in the lengthwise direction of the exterior member **30** can be adjusted when the slide regulating members **70** are not attached. Note that, also in this case, the engagement portions **63** keep the path regulating member from rotating relative to the fixing members **50** in the circumferential direction.

(59) Functions of the present embodiment will be described.

(60) With the wire harness **10** according to the present embodiment, the path regulating member **40** can be retrofitted to the outer periphery of the exterior member **30** through the insertion port **44**. In other words, the path regulating member **40** can be attached to the exterior member **30** from a direction orthogonal to the lengthwise direction of the exterior member **30**. The path regulating member **40** includes the pair of protruding portions **45** that come into contact with the outer surface of the exterior member **30**, and thus it is possible to suppress the path regulating member **40** from coming loose from the exterior member **30** through the insertion port **44**.

(61) In addition, the wire harness **10** can be fixed as a result of the path regulating member **40** attached to the exterior member **30** being held by the fixing members **50**, and the fixing members **50** being fixed to the vehicle body **V1**. At this time, the engagement portions **63** of the fixing members **50** can come into contact with the first end portion **41** and the second end portion **42** that form the insertion port **44** in the circumferential direction of the path regulating member **40**, and thus the path regulating member **40** is kept from rotating relative to the fixing members **50** in the circumferential direction.

(62) Functions of the present embodiment will be described. (1) The path regulating member **40** can be retrofitted to the outer periphery of the exterior member **30** through the insertion port **44**. In addition, the engagement portions **63** of the fixing members **50** that hold the path regulating member **40** and are fixed to the vehicle body **V1** can come into contact with the first end portion **41** and the second end portion **42** that form the insertion port **44** in the circumferential direction of the path regulating member **40**. Thus, the fixing members **50** can keep the path regulating member **40** from rotating relative to the fixing members **50** in the circumferential direction while holding the path regulating member **40**. Thus, the path regulating member **40** is kept from rotating due to vibration that occurs when the vehicle **V** is travelling, for example, and the insertion port **44** is kept from facing downward so as to oppose the ground. As a result, for example, it is possible to suppress a decrease in the ability of the path regulating member **40** to protect the exterior member **30** from flying rocks and the like, and a decrease in the durability of the wire harness **10**. In addition, the fixing members **50** that include the fixing portions **54** for fixing to the vehicle body

include the engagement portions **63** that suppress rotation of the path regulating member **40**, and thus it is possible to reduce the number of components compared with a case where their functions are realized using other members. (2) The engagement portion **63** includes the narrow portions **66** whose width decreases toward the leading end side thereof when viewed from the lengthwise direction of the path regulating member **40**, and thus the engagement portion **63** can be easily inserted into the insertion port **44**, which facilitates attachment. (3) The fixing members **50** each include the main body portion **51** and the lid portion **52** that can be fixed to the main body portion **51** and encloses and holds the path regulating member **40** in cooperation with the main body portion **51** while fixed to the main body portion **51**, and thus the fixing members **50** can be retrofitted to the path regulating member **40**. In other words, the fixing members **50** can be attached to the path regulating member **40** from a direction orthogonal to the lengthwise direction of the path regulating member **40**. (4) The engagement portion **63** protrudes from the vehicle body opposing portion **64** that opposes the attachment surface V2 of the vehicle body V1, and thus the insertion port **44** of the path regulating member **40** faces the attachment surface V2 side. Thus, for example, the insertion port **44** is kept from facing downward so as to oppose the ground. (5) Since a plurality of fixing members **50** are provided in the lengthwise direction of the path regulating member **40**, rotation of the path regulating member **40** relative to the fixing members **50** in the circumferential direction can be suppressed at a plurality of locations in the lengthwise direction of the path regulating member **40**. Thus, it is possible to more firmly suppress rotation of the path regulating member **40**. (6) The engagement portion **63** includes the protrusions **65** that enter annular recesses **32** of the exterior member **30**, and thus it is possible to suppress movement of the exterior member **30** relative to the fixing member **50** in the lengthwise direction of the exterior member **30**. (7) Movement of the path regulating member **40** relative to the exterior member in the lengthwise direction of the exterior member **30** is regulated by the slide regulating members **70**. In addition, the path regulating member **40** is allowed to move relative to the fixing members **50** in the lengthwise direction of the exterior member **30** when no slide regulating member **70** is attached, and thus it is possible to adjust the position of the path regulating member **40**. (8) The path regulating member **40** includes the protruding portions **45** that protrude from the inner surfaces of the first end portion **41** and the second end portion **42** and come into contact with the outer surface of the exterior member **30**, and thus it is possible to keep the path regulating member **40** from coming loose from the exterior member **30** through the insertion port **44**. (9) The protruding portions **45** protrude from the inner surfaces of the leading end **46** of the first end portion **41** and the leading end **47** of the second end portion **42**. Thus, for example, compared with a case where the protruding portions **45** protrude from the inner surface of the path regulating member **40** at positions slightly away from the leading end **46** of the first end portion **41** and the leading end **47** of the second end portion **42**, it is possible to further keep the path regulating member **40** from coming loose from the exterior member **30** through the insertion port **44**.

MODIFIED EXAMPLES

(63) The present embodiment can be modified and implemented as follows. The present embodiment and the modified examples below can be implemented in combination with each other as long as no technical contradictions arise. In the above embodiment, the engagement portion **63** includes the narrow portions **66** whose width decreases toward the leading end side thereof when viewed from the lengthwise direction of the path regulating member **40**, but there is no limitation thereto, and a configuration may also be adopted in which no narrow portion **66** is provided. In addition, in the above embodiment, each protrusion **65** is provided with a narrow portion **66**, but there is no limitation thereto, and, for example, the engagement portion **63** itself may configure a narrow portion whose width decreases toward the leading end side when viewed from the lengthwise direction of the path regulating member **40**. In the above embodiment, a configuration is adopted in which the fixing members **50** each include the main body portion **51** and the lid portion **52**, and can be retrofitted to the path regulating member **40**, but there is no limitation thereto, and,

for example, a configuration may also be adopted in which the lid portion **52** cannot move in a closed state. In this case, the fixing members cannot be retrofitted to the path regulating member **40**, specifically, the fixing members cannot be attached to the path regulating member **40** from a direction orthogonal to the lengthwise direction of the path regulating member **40**, and the path regulating member **40** needs to be passed through the fixing members in the lengthwise direction and attached thereto. In the above embodiment, the lid portion **52** is coupled to the main body portion **51** via the hinge portion **59**, but there is no limitation thereto, and the main body portion **51** and the lid portion **52** may be separate components. Note that, as a matter of course, even when a main body portion and a lid portion are separate components, the lid portion needs to be fixable to the main body portion, and be capable of enclosing and holding the path regulating member **40** in cooperation with the main body portion while fixed to the main body portion. In the above embodiment, a configuration is adopted in which the lid portion **52** includes the claw portion **60**, and when the lid portion **52** is brought into a closed state, the claw portion **60** is engaged with the claw engagement portion **56** of the main body portion **51**, but the lid portion **52** may be fixed to the main body portion **51** using another configuration. In the above embodiment, the engagement portion **63** protrudes from the vehicle body opposing portion **64** that opposes the attachment surface **V2** of the vehicle body **V1**, but there is no limitation thereto, and the engagement portion **63** may protrude from another section. In the above embodiment, two fixing members **50** are provided in the lengthwise direction of the path regulating member **40**, but there is no limitation thereto, and a configuration may also be adopted in which one fixing member **50** or three or more fixing members **50** are provided in the lengthwise direction of the path regulating member **40**. In the above embodiment, the engagement portion **63** includes the protrusions **65** that enter annular recesses **32** of the exterior member **30**, but there is no limitation thereto, and a configuration may also be adopted in which the protrusions **65** are not provided. In addition, the exterior member **30** is a corrugated tube in which the annular protrusions **31** and the annular recesses **32** are alternately provided in a continuous manner along the lengthwise direction, but another exterior member may be adopted. In addition, when an exterior member that does not include the annular recesses **32** is used, the protrusions **65** are unnecessary, and it is preferable that movement of the exterior member relative to the fixing members **50** in the lengthwise direction is suppressed using another configuration. In the above embodiment, the wire harness **10** includes the slide regulating members **70** for regulating movement of the path regulating member **40** relative to the exterior member **30** in the lengthwise direction of the exterior member **30**, but there is no limitation thereto, and a configuration may also be adopted in which the slide regulating members **70** are not provided. In addition, each slide regulating member **70** is a piece of adhesive tape, but a slide regulating member other than an adhesive tape may also be used. In the above embodiment, a configuration is adopted in which the path regulating member **40** is allowed to move relative to the fixing members **50** in the lengthwise direction of the exterior member **30**, in a state where no slide regulating member **70** is attached, but there is no limitation thereto, and a configuration may also be adopted in which the path regulating member **40** is not allowed to move relative to the fixing members **50** in the lengthwise direction of the exterior member **30**. That is to say, a configuration may also be adopted in which each fixing member **50** and the path regulating member **40** include sections that are directly engaged with each other in the lengthwise direction of the path regulating member **40**. In the above embodiment, the path regulating member **40** includes the protruding portions **45** that protrude from the inner surfaces of the first end portion **41** and the second end portion **42** and come into contact with the outer surface of the exterior member **30**, but there is no limitation thereto, and a configuration may also be adopted in which no protruding portions **45** are provided. In addition, the protruding portions **45** may protrude from the inner surface (of the path regulating member **40**) at positions slightly away from the leading end **46** of the first end portion **41** and the leading end **47** of the second end portion **42**. In the above embodiment, the fixing members **50** are fixed to the vehicle body **V1** by the fixing pieces **62** that extend from the attachment surface **V2** toward the

ground, but there is no limitation thereto, and the fixing members **50** may be fixed using another configuration. A configuration may be adopted in which, for example, the fixing members **50** are fixed to the attachment surface **V2** via other attachment brackets. In addition, for example, a configuration may also be adopted in which each fixing member **50** includes a fitting portion that is fitted into an attachment hole formed in the attachment surface **V2**, and the fixing member **50** is fixed to the attachment surface **V2** by fitting the fitting portion into the attachment hole. The path regulating member **40** may be made of metal. The path regulating member **40** may be made of a metal material that is iron-based, copper-based, aluminum-based, or the like. With such a configuration, for example, when the path regulating member **40** is disposed at a position near a heat source of the vehicle, it is possible to suppress an increase in temperature inside the exterior member **30**, and accordingly, an increase in the temperature of the electric wire member **20**. The exterior member **30** may be formed by a metal layer containing a metal material provided on the outer surface of a corrugated tube. Such a metal layer can be provided through plating, for example. It is preferable that the metal layer is provided over all outer surfaces of the annular protrusions **31** and the annular recesses **32** of the corrugated tube. A metal material that has a small radiation factor such as aluminum is preferably used for the outermost surface of the metal layer, for example. With such a configuration, for example, when the exterior member **30** is disposed at a position near a heat source of the vehicle, it is possible to suppress an increase in temperature inside the exterior member **30**, and accordingly, an increase in the temperature of the electric wire member **20**. The exterior member **30** may include a slit that extends along the lengthwise direction of the exterior member **30** over the entire length of the exterior member **30**. In this case, the exterior member **30** is preferably sealed over the entirety in the circumferential direction, for example, by winding tape around the outer periphery of the exterior member **30** so as to block the slit over the entire length thereof. Accordingly, it is possible to suppress a decrease in the water blocking properties of the exterior member that includes the slit. The electric wire member **20** may include one electric wire **21**, or may include three or more electric wires **21**. The braided member **24** may be omitted from the electric wire member **20**. The wire harness **10** may include a plurality of path regulating members **40** spaced apart from each other in the lengthwise direction of the exterior member **30**. The path regulating member **40** is not limited to a path regulating member provided under the floor of the vehicle **V**. The path regulating member **40** may be provided in the vehicle interior of the vehicle **V**, as long as it is provided at a portion of the routing path of the electric wire member **20** that extends linearly, for example. As shown in FIG. 5, the leading end **46** of the first end portion **41** and the leading end **47** of the second end portion **42** may oppose two end portions in the circumferential direction of the engagement portion **63**. In a state where the path regulating member **40** is held by the fixing members **50**, a gap in the circumferential direction may be formed between the first end portion **41** and the engagement portion **63** and between the second end portion **42** and the engagement portion **63**. As shown in FIG. 2, the surface of the leading end **46** of the first end portion **41** may have an arc shape that is smoothly continuous with the surface of the protruding portion **45** of the first end portion **41** when viewed from the lengthwise direction of the path regulating member **40**. In other words, the leading end **46** and the protruding portion **45** of the first end portion **41** may have arc-shaped surfaces that extend on the same circumference when viewed from the lengthwise direction of the path regulating member **40**. Similarly, the surface of the leading end **47** of the second end portion **42** may have an arc shape that is smoothly continuous with the surface of the protruding portion **45** of the second end portion **42**. In other words, the leading end **47** and the protruding portions **45** of the second end portion **42** may have arc-shaped surfaces that extend on the same circumference when viewed from the lengthwise direction of the path regulating member **40**. Note that, the first end portion **41** may be referred to as a “first edge portion”, and the second end portion **42** may be referred to as a “second edge portion”. In addition, one of the pair of protruding portions **45** may be referred to as a “first protruding portion”, and the other of the pair of protruding portions **45** may be referred to as a “second protruding portion”. As

shown in FIGS. 2 and 3, the insertion port 44 is a groove that extends linearly along the lengthwise direction of the path regulating member 40 over the entire length of the path regulating member 40, and the groove may be open at two ends in the lengthwise direction of the path regulating member 40. In a state where the path regulating member 40 is held by the fixing members 50, the entire insertion port 44 may face a side on which the attachment surface V2 is provided, for example, the side opposite to the ground. The path regulating member 40 may cover the exterior member 30 from the opposite side to the attachment surface V2, for example, the ground side. The path regulating member 40 may be referred to as a “path defining member” that defines a path of the electric wire member 20. As shown in FIGS. 3 and 5, the engagement portion 63 may protrude from the vehicle body opposing portion 64 on the side opposite to the attachment surface V2. The leading end surface of the engagement portion 63 may have an arc shape disposed on the same circumference as an arc defined by the inner surface of the coupling portion 43 when viewed from the lengthwise direction of the path regulating member 40. The leading end surface of each protrusion 65 has an arc shape that forms a portion of a circle that is concentric with an arc defined by the leading end surface of the engagement portion 63. The thickness in the radial direction of the coupling portion 43 may be constant in the circumferential direction. As shown in FIG. 6, in a state where the path regulating member 40 is held by the fixing members 50, at least a portion of the leading end surface of the protrusions 65 may come into contact with the outer peripheral surface of the annular protrusions 32.

(64) As shown in FIG. 6, each fixing member 50 may include a plurality of engagement portions 63 that are disposed with a space therebetween in the lengthwise direction of the path regulating member 40. Narrow portions 65 may be provided at an interval (pitch) that is the same as the interval at which the annular recesses 32 are provided, or an interval of an integer multiple thereof, in the lengthwise direction of the path regulating member 40. The width along the lengthwise direction of each narrow portion 65 may be smaller than the width of an annular recess 32, in other words, the distance between adjacent annular protrusions 31. The present disclosure encompasses the following aspects. Reference signs of several constituent elements of exemplary embodiments are added to aid in comprehension, not for limitation. Some of the items described in the following aspects may be omitted, and some of the items described in the aspects may be selected or extracted and combined. [Supplementary Note 1] In one aspect of the present disclosure, the protruding portions (45) respectively protrude from the inner surfaces of the leading end of the first end portion (41) and the leading end of the second end portion (42), and when viewed in the lengthwise direction of the path regulating member (40), the surface of the leading end (46) of the first end portion (41) may have an arc shape smoothly continuous with the surface of the protruding portion (45), and the surface of the leading end (47) of the second end portion (42) may have an arc shape smoothly continuous with the surface of the protruding portion (45).

[Supplementary Note 2] In one aspect of the present disclosure, the leading end surface of the engagement portion (63) has an arc shape disposed on the same circumference of an arc defined by the inner surface of the coupling portion (43), the leading end surface of the protrusion (65) has an arc shape that forms a portion of a circle that is concentric with an arc defined by the leading end surface of the engagement portion (63), and at least a portion of the leading end surface of the protrusion (65) may come into contact with the outer peripheral surface of the annular protrusion (32) in a state where the path regulating member (40) is held by the fixing members (50).

[Supplementary Note 3] In one aspect of the present disclosure, the fixing member (50) may include a plurality of engagement portions (63) that are spaced apart in the lengthwise direction of the path regulating member (40). [Supplementary Note 4] In one aspect of the present disclosure, the narrow portions (65) may be provided at an interval that is the same as the interval at which the annular recesses 32 are provided, or an interval of an integer multiple thereof, in the lengthwise direction of the path regulating member (40).

Claims

1. A wire harness comprising: an electric wire; an exterior tube that covers an outer periphery of the electric wire; a path regulator that extends along a lengthwise direction of the exterior tube while covering a circumferential portion of an outer periphery of the exterior tube, and regulates a path along which the electric wire is routed; a fixing member that holds the path regulator and is fixed to a vehicle body; and a slide regulator that regulates movement of the path regulator relative to the exterior tube in the lengthwise direction of the exterior tube, wherein the path regulator includes: an insertion port that is an opening that extends along a lengthwise direction of the path regulator over an entire length of the path regulator and is configured to enable the exterior tube to be inserted thereinto, and a first end and a second end that are positioned on opposite sides to each other in a circumferential direction of the path regulator, and from the insertion port, the fixing member includes an engagement that protrudes toward the insertion port, the engagement is configured to abut at least one of the first end and the second end and stop rotation of the path regulator in the circumferential direction of the path regulator, and the path regulator is allowed to move relative to the fixing member in the lengthwise direction of the exterior tube in a case where the slide regulator is not attached.
 2. The wire harness according to claim 1, wherein the engagement includes a narrow portion whose width decreases toward a leading end side of the engagement when viewed from the lengthwise direction of the path regulator.
 3. The wire harness according to claim 1, wherein the fixing member includes: a main body, and a lid that is fixable to the main body, and encloses and holds the path regulator in cooperation with the main body while fixed to the main body.
 4. The wire harness according to claim 1, wherein the engagement protrudes from a vehicle body opposing portion that opposes an attachment surface of the vehicle body.
 5. The wire harness according to claim 1, wherein the fixing member is one of a plurality of fixing members provided in the lengthwise direction of the path regulator.
 6. The wire harness according to claim 1, wherein: the exterior tube is a corrugated tube that has a bellows structure in which an annular protrusion and an annular recess are alternately provided in a continuous manner along the lengthwise direction of the exterior tube, and the engagement includes a protrusion that enters the annular recess.
 7. The wire harness according to claim 1, wherein the path regulator includes a first protrusion that protrudes from an inner surface of the first end and a second protrusion that protrudes from an inner surface of the second end toward the exterior tube, and both come into contact with an outer surface of the exterior tube.
 8. The wire harness according to claim 7, wherein the first protrusion and the second protrusion protrude, respectively, from an inner surface of a leading end of the first end and a leading end of the second end.
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